

Chapter 3 — Operational Amplifier Circuits and Oscillators

Electronics Device and Circuit (ENEX 151 / EX 151) · Tribhuvan University, IOE · BE (BEI, BCT) · Year I / Part II

Exam-oriented notes with embedded Past-Year Questions (2081 Ashwin R · 2082 Baishakh B · 2082 Bhadra R · 2083 Baishakh B)

References: Sedra-Smith (oscillator loop-gain method, current mirrors, differential pair, active loads), Boylestad (oscillator circuits), Floyd (555 timer), Millman-Halkias (feedback treatment)

Figures: 13 diagrams rendered locally — response curves computed from the actual network equations (fetch report: web download unavailable, 0/13 from web)

How to Use These Notes / Chapter Overview

This chapter has two threads that the exam always samples together for **exactly 13 marks**. Thread one: **oscillators** — an amplifier plus a frequency-selective positive-feedback network satisfying the **Barkhausen criterion**; the network decides the family (RC for audio, LC for radio, crystal for precision) and the frequency formula. The 555 timer joins as a *relaxation* oscillator that works by charging a capacitor between two thresholds instead of by resonance. Thread two: the **IC building blocks** that make analog chips possible — current mirrors (simple → base-compensated → Widlar), the differential pair, and the active load that doubles its gain.

PYQ Frequency Analysis — Chapter 3 is worth exactly 13 marks every single year

Topic	2081 Ashwin	2082 Baishakh	2082 Bhadra	2083 Baishakh	Verdict
One sinusoidal oscillator (circuit + operation + frequency)	✓ Colpitts [4]	✓ Hartley [4]	✓ RC phase-shift [6]	✓ Wien bridge [1+5]	★ 4/4 GUARANTEED — type rotates
Barkhausen criterion (lead-in of the oscillator Q)	✓ stated	implied	✓ stated	implied	attached to above
Current source (Widlar / mirror compensation)	✓ Widlar R_{out} [5]	–	✓ Widlar op.+adv. [5+2]	✓ compensated mirror [5+2]	★ 3/4 near-guaranteed
555 astable frequency derivation	✓ [4]	✓ [4]	–	–	2/4
Differential amplifier with active load (gain $\times 2$ proof)	–	✓ [5]	–	–	1/4
Precision rectifiers / crystal osc. / square-triangle gen.	–	–	–	–	never asked — readiness only

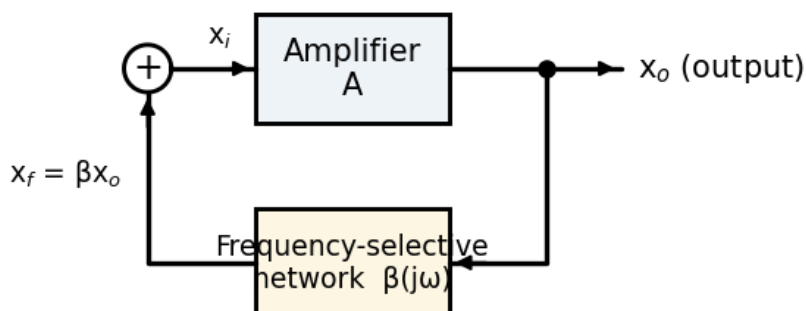
Strategic conclusions

- ★ The four sinusoidal oscillators have each appeared **exactly once** (Colpitts → Hartley → RC phase-shift → Wien). The rotation is complete, so the next paper repeats one or reaches for the crystal oscillator — prepare **all four** plus the crystal paragraph. Every answer has the same skeleton: *Barkhausen statement* → *circuit* → *how the phase condition is met* → *frequency derivation* → *gain condition*.
- ★ The current-source question missed only one paper. Prepare the trio as one story: simple mirror (error $2/\beta$) → base-compensated (error $2/\beta^2$) → Widlar (microamp currents + huge R_{out}). Two of the three PYQs demand the *derivations*, not just circuits.
- The 555 derivation is four marks of pure algebra on one exponential-charging law — the cheapest derivation in the syllabus; both PYQs are word-for-word identical.

3.1 Basic Principles of Sinusoidal Oscillators

◆ Intuition first

An oscillator is an amplifier that has been given a whisper of its own output and told to trust it. Connect the output back to the input through a network that (a) returns the signal *exactly in phase* and (b) preserves *exactly* enough amplitude, and any noise component at the one frequency satisfying both conditions will circulate forever: DC power in, sine wave out, no input required. The feedback network is frequency-selective precisely so that only ONE frequency wins.



no external input — the loop feeds itself: $x_f = x_i \Rightarrow A\beta = 1$

Fig. 1 — Positive-feedback loop with no external input. Self-sustainment requires the returned signal to equal the required input. [rendered locally]

Formal statement + one-line derivation — the Barkhausen criterion (asked verbatim: 2082 Bhadra Q5, 2081 Q6)

Let the amplifier gain be A and the feedback factor $\beta(j\omega)$. The loop sustains itself when the fed-back signal reproduces the amplifier input: $x_f = x_i$. Since $x_f = \beta \cdot x_o = \beta \cdot A \cdot x_i$, this requires **$A \cdot \beta = 1$** — the loop gain must be exactly unity as a complex number, i.e.:

1. Magnitude condition: $|A \cdot \beta(j\omega_0)| = 1$ **2. Phase condition:** $\angle A \cdot \beta(j\omega_0) = 0^\circ$ (or $n \cdot 360^\circ$)

- The **phase condition selects the frequency**: only ω_0 makes the loop phase zero (see Figs. 3 and 5 — the phase curves cross the critical value at a single frequency).
- The **magnitude condition sets the minimum amplifier gain**: the amplifier must exactly replace what the network attenuates.
- **Start-up (worth a mark)**: design with $|A\beta|$ slightly > 1 so oscillation grows out of noise; a nonlinearity (device saturation, lamp/diode limiter, AGC) then trims the effective gain back to exactly 1 at the desired amplitude.
- ⚠ Common mistake: writing $|A\beta| > 1$ as the *steady-state* criterion — that describes start-up only; sustained sinusoidal oscillation is $|A\beta| = 1$.

3.2 Op-Amp Multivibrators and RC Oscillator Circuits

Square / triangular generation (readiness paragraph — never asked)

● **LOW** An op-amp astable (Schmitt trigger with an RC in negative feedback) snaps between saturation rails, generating a square wave of period $T = 2RC \cdot \ln[(1+\beta)/(1-\beta)]$ where $\beta = R_1/(R_1+R_2)$ is the hysteresis fraction; feeding the square wave through an op-amp **integrator** yields the triangular wave. Two-op-amp function-generator loop: comparator + integrator.

A. RC Phase-Shift Oscillator ★ REPEATED PYQ TOPIC — PYQ 2082 Bhadra Q5 [6, includes Barkhausen]

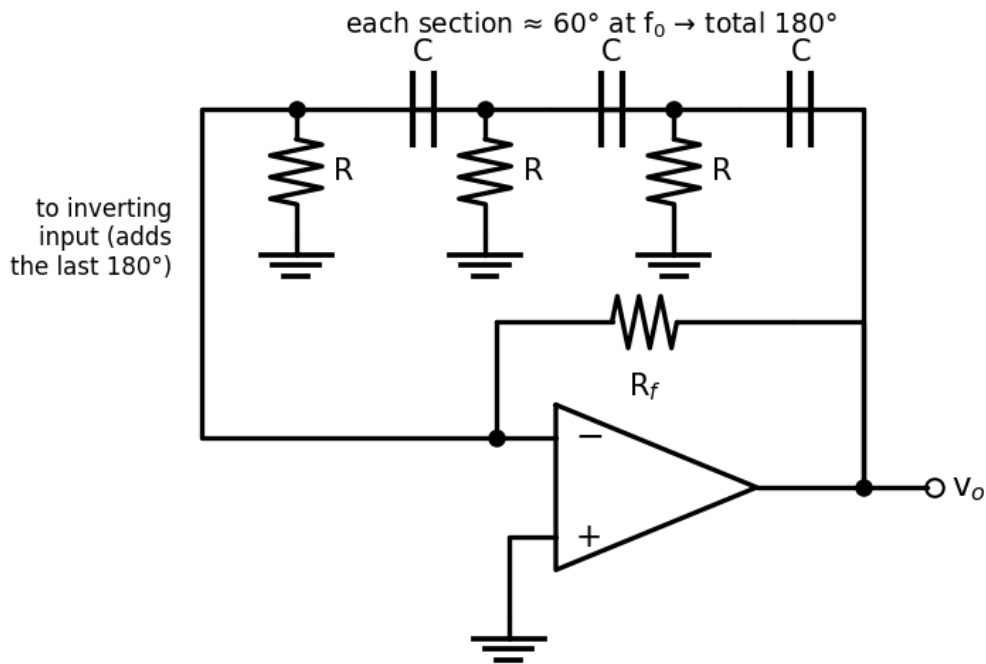


Fig. 2 — RC phase-shift oscillator (op-amp version): inverting amplifier (-180°) + three cascaded C-R sections (-180° total at f_0). Magnitude condition demands $|R_f/R| \geq 29$. BJT version: one CE stage plays the inverting amplifier. [rendered locally]

Operation: the inverting amplifier contributes a fixed 180° ; each identical C-R section contributes about 60° of leading phase at the oscillation frequency; three of them add the remaining $180^\circ \rightarrow$ total loop phase $360^\circ \checkmark$. The magnitude penalty of pushing a signal through three attenuating sections is a factor of 29, which the amplifier must repay.

Frequency derivation (write it as this chain)

- 1. Nodal analysis of the three-section ladder (each series C, shunt R) gives the feedback transfer function $\beta(j\omega) = v_f/v_o = 1 / [(1 - 5\alpha^2) + j\alpha(6 - \alpha^2)]$, where $\alpha = 1/(\omega RC)$. (Quote this standard result; deriving it is three KCL equations — offer if time allows.)
- 2. Phase condition $\Rightarrow$ the imaginary part must vanish: $\alpha(6 - \alpha^2) = 0 \Rightarrow \alpha^2 = 6 \Rightarrow \omega RC = 1/\sqrt{6}$.

$$f_0 = 1/(2\pi RC\sqrt{6})$$

- 3. At $\alpha^2 = 6$ the real part is $1 - 5(6) = -29$, so $\beta = -1/29$: the network attenuates $29\times$ and inverts (its -180°). Magnitude condition $\Rightarrow |A| \geq 29$ (op-amp: $R_f/R \geq 29$; BJT: h_{fe} comfortably above ≈ 45 in practice).
- Refinement to mention: the BJT version loads the ladder, shifting $f_0 = 1/[2\pi RC\sqrt{(6 + 4R_C/R)}]$.

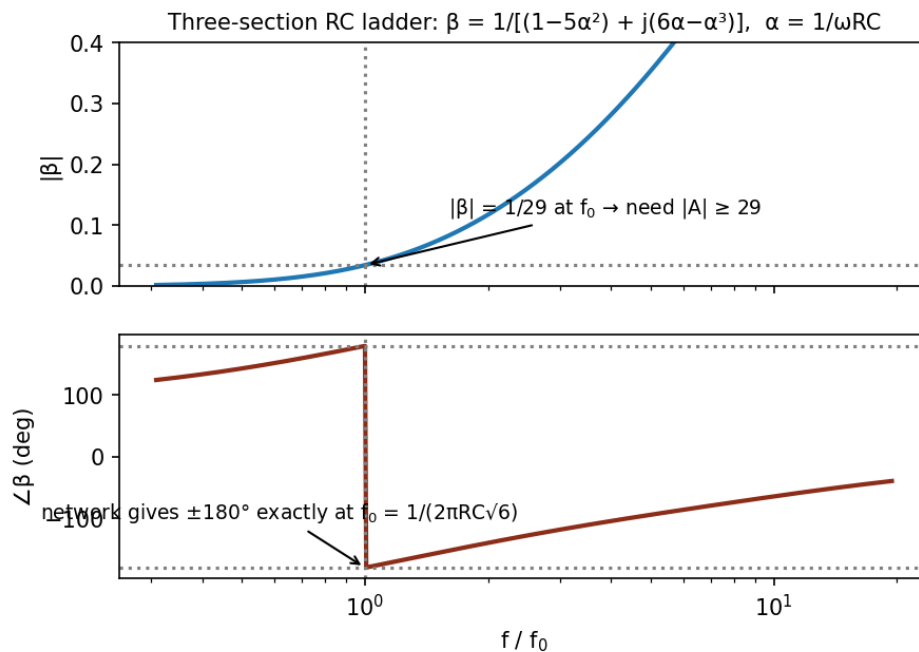


Fig. 3 — Computed $\beta(j\omega)$ of the three-section ladder: $|\beta| = 1/29$ exactly where the phase reaches $\pm 180^\circ$ — the two Barkhausen conditions meeting at f_0 . [computed from the network equation]

- $\triangle$ Mistakes: forgetting the amplifier's own 180° (claiming the network alone gives 360°) · quoting f_0 with $\sqrt{6}$ in the numerator · each section is $\approx 60^\circ$ only at f_0 , not at all frequencies.

B. Wien Bridge Oscillator ★ REPEATED PYQ TOPIC — PYQ 2083 Baishakh Q5 [1+5]

Part (a) [1] — RC vs LC oscillators by application: RC oscillators serve **low/audio frequencies** (a few Hz to a few hundred kHz) where LC would need impractically large, lossy, drift-prone inductors; LC oscillators serve **high/RF frequencies** (hundreds of kHz to GHz) where small high-Q tanks are easy and RC phase networks become inaccurate. Give both halves for the mark.

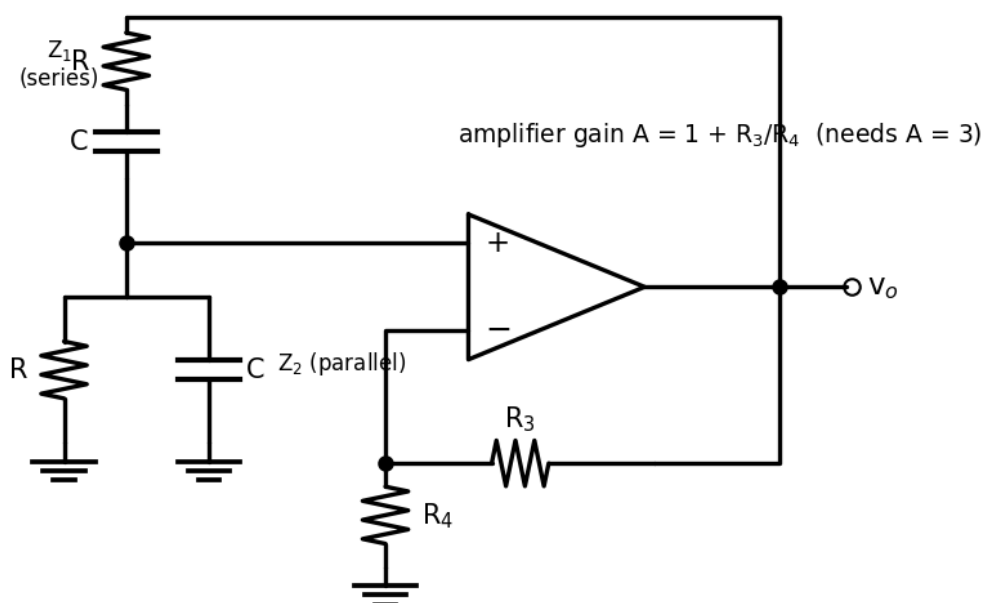


Fig. 4 — Wien bridge oscillator: non-inverting amplifier (gain $1 + R_3/R_4$) with the series-RC (Z_1) / parallel-RC (Z_2) divider as positive feedback. The four arms Z_1 , Z_2 , R_3 , R_4 form the 'bridge'. [rendered locally]

Operation + full derivation [5]

- **1. Impedances:** $Z_1 = R + 1/j\omega C$ (series arm), $Z_2 = R/(1 + j\omega RC)$ (parallel arm).

- **2. Feedback factor:** $\beta = Z_2/(Z_1 + Z_2) = 1/(1 + Z_1/Z_2)$. Compute $Z_1/Z_2 = (R + 1/j\omega C)(1 + j\omega RC)/R = 1 + j\omega RC + 1/(j\omega RC) + 1 = 2 + j(\omega RC - 1/\omega RC)$. Therefore:

$$\beta(j\omega) = 1 / [3 + j(\omega RC - 1/\omega RC)]$$

- **3. Phase condition:** imaginary part zero $\Rightarrow \omega RC = 1/(\omega RC) \Rightarrow \omega_0 = 1/RC$:

$$f_0 = 1/(2\pi RC) \quad \text{and at } f_0: \beta = 1/3 \text{ (its maximum, purely real — see Fig. 5)}$$

- **4. Magnitude condition:** the amplifier must supply gain 3: $A = 1 + R_3/R_4 = 3 \Rightarrow R_3 = 2R_4$. Design with R_3 marginally above $2R_4$ for start-up, and stabilise amplitude with an incandescent lamp or diode limiter in the R_3 - R_4 arm — mentioning this earns the polish mark.

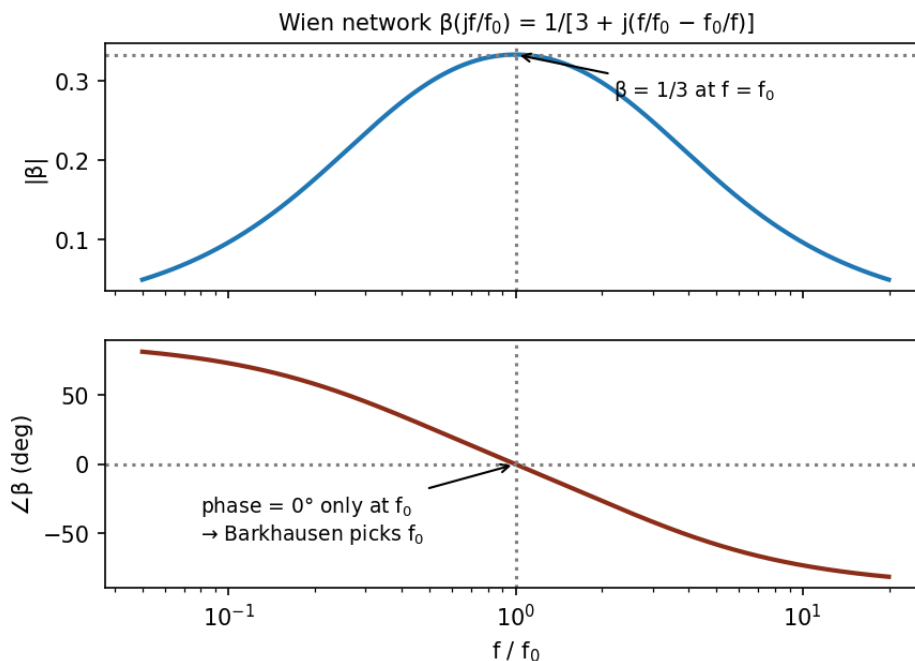


Fig. 5 — Computed Wien-network response: $|\beta|$ peaks at exactly $1/3$ and the phase crosses 0° only at f_0 — the network is a gentle band-pass whose centre the Barkhausen phase condition selects. [computed from the network equation]

- $\triangle$ Mistakes: swapping the series/parallel arms · $A = 3$ means $1 + R_3/R_4 = 3$, i.e. $R_3 = 2R_4$ (not $R_3 = 3R_4$) · $\beta = 1/3$ at f_0 , not $1/29$ (that's the phase-shift ladder).

3.3 LC and Crystal Oscillators

A. Colpitts Oscillator — PYQ 2081 Ashwin Q6 [4, includes Barkhausen]

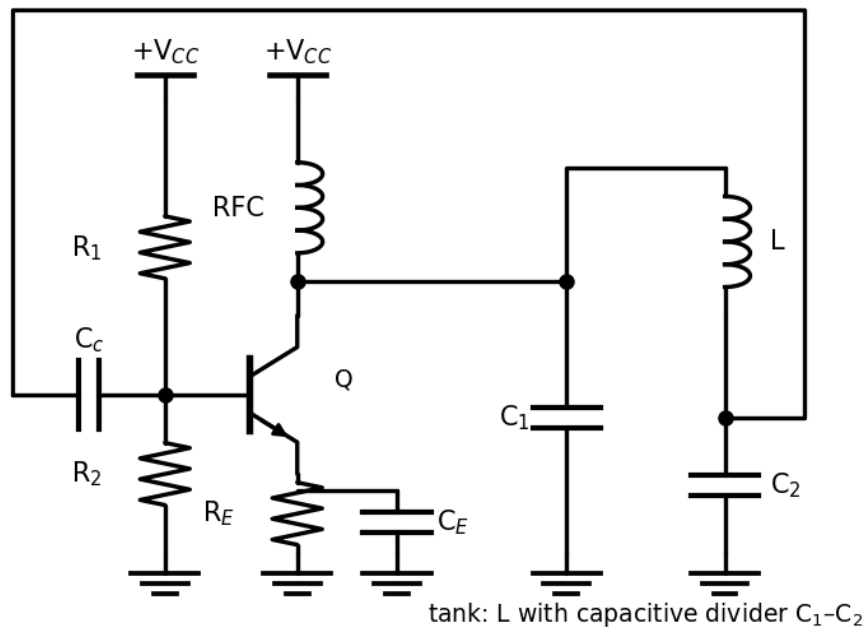


Fig. 6 — Colpitts oscillator: CE amplifier with an LC tank whose CAPACITIVE divider C_1 - C_2 (junction grounded) provides the feedback tap; RFC feeds DC while blocking RF. [rendered locally]

- **Operation:** the tank rings at its resonant frequency; the C_1 - C_2 divider returns a fraction of the tank voltage to the base. The CE stage inverts (180°) and the divider's ground-referenced tap inverts again (180°) $\rightarrow$ loop phase 360° ✓ at resonance.
- **Frequency derivation:** around the tank loop L - C_1 - C_2 the net reactance must vanish at resonance:
 $\omega_0 L = 1/(\omega_0 C_1) + 1/(\omega_0 C_2) \Rightarrow \omega_0^2 = (1/L) \cdot (1/C_1 + 1/C_2) = 1/(L \cdot C_{eq})$ with the two capacitors in series:

$$C_{eq} = C_1 C_2 / (C_1 + C_2) \quad f_0 = 1/(2\pi\sqrt{L \cdot C_{eq}})$$

- **Gain condition:** the same tank current flows through both capacitors, so the voltages divide as $v_{C2}/v_{C1} = C_1/C_2 \Rightarrow \beta = C_1/C_2$ and the amplifier must give $|A| \geq C_2/C_1$ (BJT: $h_{fe} \geq C_2/C_1$).
- **Memory hook:** Colpitts = Capacitive divider; Hartley = inductive (H has two vertical strokes $\approx$ two Ls).

B. Hartley Oscillator — PYQ 2082 Baishakh Q6 [4]: draw the circuit and derive its frequency

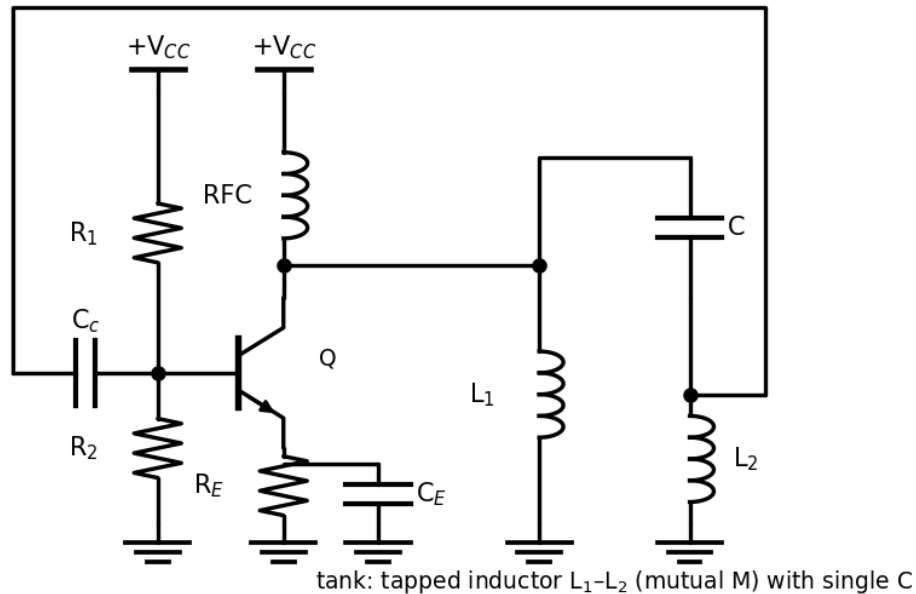


Fig. 7 — Hartley oscillator: identical skeleton, but the tank is a TAPPED INDUCTOR L_1 - L_2 (tap grounded, mutual inductance M) with a single capacitor C across the whole coil. [rendered locally]

- **Frequency derivation:** at resonance the loop reactance is zero: $\omega_0(L_1 + L_2 + 2M) = 1/(\omega_0 C)$ — the two coil halves are in series and their mutual flux adds (aiding connection), so:

$$L_{eq} = L_1 + L_2 + 2M \quad f_0 = 1/(2\pi\sqrt{L_{eq} \cdot C})$$

- **Gain condition:** the coil halves carry the same tank current, so voltages divide as the (mutual-corrected) inductances: $\beta = (L_2 + M)/(L_1 + M) \approx L_2/L_1 \Rightarrow$ amplifier gain $\geq (L_1 + M)/(L_2 + M) \approx L_1/L_2$ (labels per Fig. 7).
- Δ Mistakes (both LC types): forgetting the $+2M$ term · writing $C_{eq} = C_1 + C_2$ (that is the *parallel* formula — here the tank current passes through them in *series*) · omitting the RFC's role (feeds DC bias, isolates the tank from the supply).

C. Crystal oscillators (readiness paragraph — the likely fresh question)

● **MEDIUM** A quartz crystal is an electromechanical resonator: electrically it behaves as a series R-L-C (the 'motional' arm, with an enormous equivalent L and tiny C) in parallel with the electrode capacitance C_0 . It shows a **series resonance** $f_s = 1/(2\pi\sqrt{L_s C_s})$ and a parallel antiresonance just above it, with $Q \approx 10^4$ - 10^6 — thousands of times better than any LC tank — giving ppm-level frequency stability against temperature and supply. Used in place of the tank (e.g. Pierce configuration) wherever accuracy matters: clocks, microprocessors, RF carriers.

RC vs LC vs Crystal — the comparison table

Property	RC (phase-shift, Wien)	LC (Colpitts, Hartley)	Crystal
Frequency range	≈ 1 Hz – few 100 kHz (audio)	≈ 100 kHz – GHz (RF)	≈ 10 kHz – 100s of MHz
Frequency-setting element	R-C phase network	resonant tank	quartz mechanical resonance
Q / selectivity	low	moderate (10–100s)	extreme (10^4 – 10^6)
Stability	fair (R, C drift)	good	excellent (ppm)
Typical use	audio generators, function gens	RF transmitters, mixers, tuners	clocks, carriers, references

3.4 Integrated-Circuit Timers — the 555 Astable ★ REPEATED PYQ TOPIC

Asked twice, word-for-word: 2081 Ashwin Q7 [4] and 2082 Baishakh Q11 [4]: “Derive the frequency of oscillation of a 555 timer astable multivibrator.” The circuit + waveforms + the two-line exponential derivation IS the full answer.

◆ Intuition

No resonance here: the 555 is a **relaxation** oscillator. A capacitor shuttles between two fixed thresholds — $\frac{1}{3}V_{CC}$ and $\frac{2}{3}V_{CC}$, set by an internal 5k-5k-5k divider — while two comparators and a flip-flop steer it: charge up through $R_A + R_B$, hit the top threshold, discharge through R_B alone into pin 7, hit the bottom threshold, repeat. The output square wave is simply the flip-flop's state.

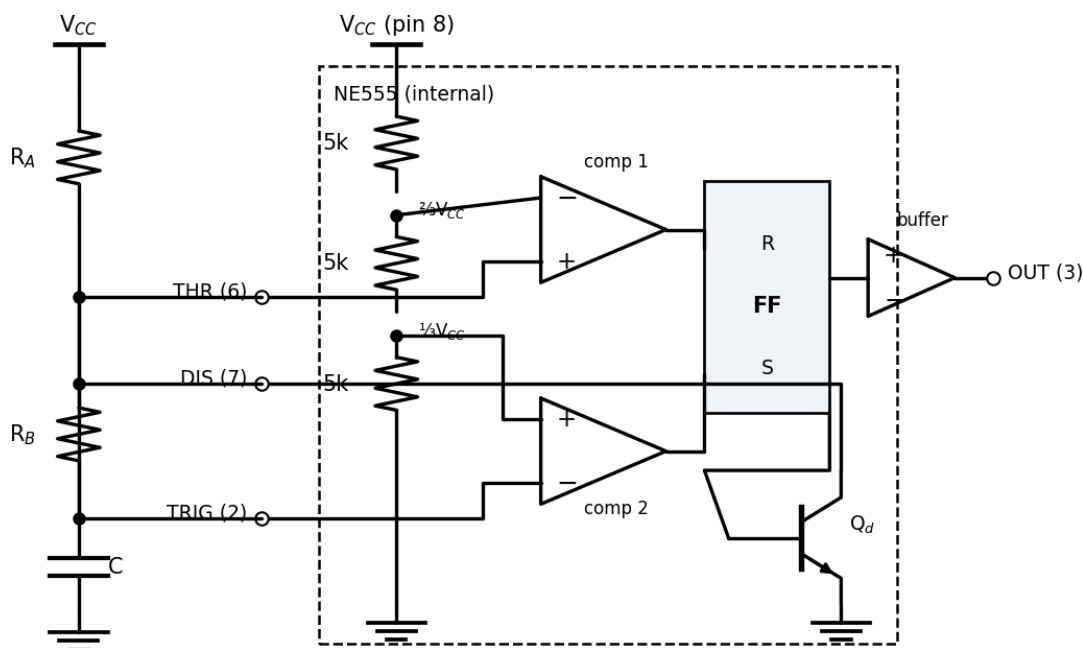


Fig. 8 — 555 astable: internal divider, two comparators, flip-flop, discharge transistor Q_d — with the external R_A , R_B , C wiring (pins 2 and 6 tied). [rendered locally]

Operation (narrate the loop once — 1 mark)

- v_C rising through $R_A + R_B$; at $v_C = \frac{2}{3}V_{CC}$ comparator 1 **Resets** the FF → output LOW, Q_d ON → C discharges through R_B into pin 7.
- At $v_C = \frac{1}{3}V_{CC}$ comparator 2 **Sets** the FF → output HIGH, Q_d OFF → charging resumes. Repeat forever.

The derivation (the remaining 3 marks — every line matters)

Universal RC step response toward a target: $v(t) = V_{\text{final}} + (V_{\text{initial}} - V_{\text{final}}) \cdot e^{-t/\tau}$.

- **Charging (output HIGH):** $V_{\text{initial}} = \frac{1}{3}V_{CC}$, $V_{\text{final}} = V_{CC}$, $\tau_1 = (R_A + R_B)C$. Set $v(t_H) = \frac{2}{3}V_{CC}$: $\frac{2}{3}V_{CC} = V_{CC} - \frac{2}{3}V_{CC} \cdot e^{-t_H/\tau_1} \Rightarrow e^{-t_H/\tau_1} = \frac{1}{2} \Rightarrow t_H = \tau_1 \cdot \ln 2 = \mathbf{0.693(R_A + R_B)C}$
- **Discharging (output LOW):** $V_{\text{initial}} = \frac{2}{3}V_{CC}$, $V_{\text{final}} = 0$, $\tau_2 = R_B C$. Set $v(t_L) = \frac{1}{3}V_{CC}$: $\frac{1}{3}V_{CC} = \frac{2}{3}V_{CC} \cdot e^{-t_L/\tau_2} \Rightarrow e^{-t_L/\tau_2} = \frac{1}{2} \Rightarrow t_L = \mathbf{0.693 \cdot R_B C}$

$$T = t_H + t_L = \mathbf{0.693(R_A + 2R_B)C} \quad f = 1/T \approx \mathbf{1.44 / [(R_A + 2R_B)C]} \quad \text{Duty cycle } D = t_H/T = \mathbf{(R_A + R_B)/(R_A + 2R_B) > 50\%}$$

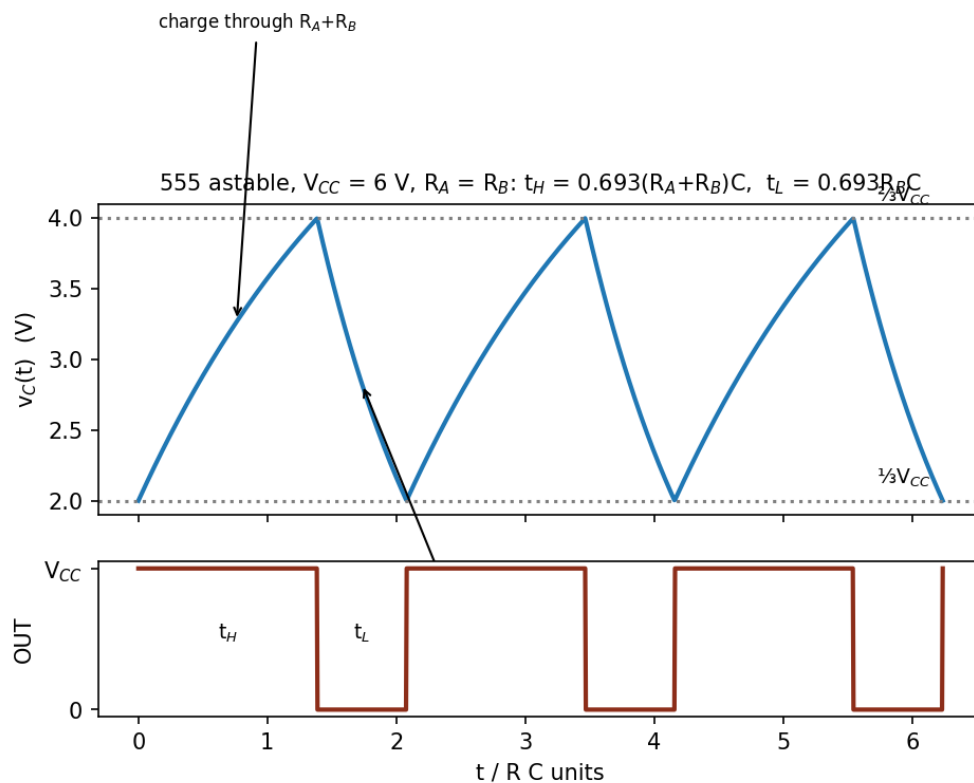


Fig. 9 — Computed 555 waveforms: $v_C(t)$ shuttling exponentially between the two thresholds, and the output square wave (drawn for $R_A = R_B$, giving $D = 2/3$). [computed from the exponential equations]

- **Closing remarks worth marks:** both thresholds scale with V_{CC} , so **f is independent of the supply voltage**; D is always $>50\%$ in this basic circuit (a diode across R_B fixes 50%); monostable cousin: $T = 1.1RC$ (one-liner readiness).
- $\triangle$ Mistakes: using $R_A + R_B$ for the DISCHARGE too (discharge sees only R_B — pin 7 bypasses R_A) · $1.44 = 1/0.693$ belongs in f , not in T · thresholds are $\frac{1}{3}$ and $\frac{2}{3}$ of V_{CC} , not fixed voltages.

3.5 Precision Rectifier Circuits

● **LOW** (Never asked; hold 3–4 mark readiness.) An ordinary diode cannot rectify signals smaller than ≈ 0.7 V. Place the diode **inside an op-amp's negative-feedback loop** and the op-amp supplies the 0.7 V itself: the effective turn-on error shrinks by the open-loop gain ($0.7/A \approx \mu\text{V}$) — the **superdiode**. The improved half-wave version adds a second diode to keep the op-amp out of saturation; a full-wave / absolute-value circuit = precision half-wave stage + weighted summing amplifier. Applications: AC millivoltmeters, peak detectors, envelope detectors — anywhere the signal is smaller than a diode drop.

3.6 Bias Circuits for IC Design — Current Mirrors ★ REPEATED PYQ TOPIC

◆ Why mirrors exist

On a chip, resistors and capacitors are expensive in area and inaccurate in value, but **matched transistors are nearly free and nearly identical**. So IC designers generate ONE reference current with one resistor, then *copy* it everywhere: two transistors sharing the same V_{BE} conduct the same collector current. The only error is the base current stolen from the reference — and the whole art is making that theft smaller.

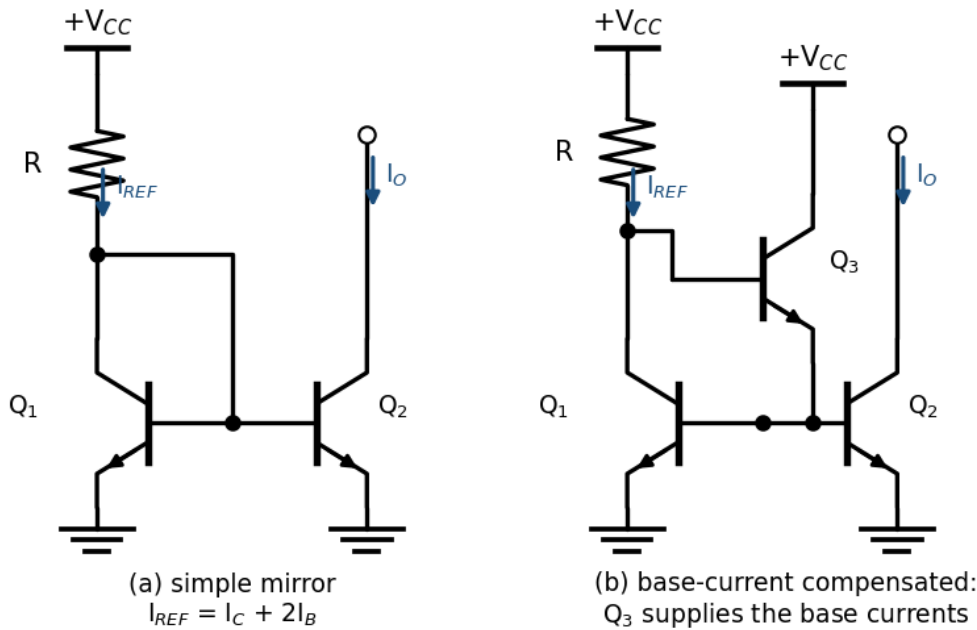


Fig. 10 — (a) Simple two-transistor mirror — Q_1 diode-connected. (b) Base-current-compensated mirror — emitter-follower Q_3 supplies both base currents, stealing only their $(\beta+1)$ -th part from the reference node. [rendered locally]

Simple mirror — derivation (write every line)

- 1. Q_1 is diode-connected; R sets the reference: $I_{REF} = (V_{CC} - V_{BE})/R$.
- 2. Matched devices, same $V_{BE} \Rightarrow I_{C1} = I_{C2} = I_C$.
- 3. KCL at the reference node: $I_{REF} = I_{C1} + I_{B1} + I_{B2} = I_C + 2I_C/\beta = I_C(1 + 2/\beta)$.

$$I_O = I_{REF} / (1 + 2/\beta) \quad \text{fractional error} \approx 2/\beta \quad (2\% \text{ at } \beta = 100)$$

★ Base-current compensation — PYQ 2083 Baishakh Q6 [5+2]: explain how it reduces the I_O - I_{REF} difference

- [5] Derivation.** Insert emitter-follower Q_3 between Q_1 's collector and the common bases (Fig. 10b). The two base currents $2I_B = 2I_C/\beta$ are now delivered by Q_3 's emitter; Q_3 's base draws only their $(\beta+1)$ -th part from the reference node: $I_{B3} = 2I_C/[\beta(\beta+1)]$. KCL at the reference node: $I_{REF} = I_C + 2I_C/[\beta(\beta+1)] = I_C[1 + 2/(\beta^2 + \beta)]$. Hence:

$$I_O = I_{REF} / [1 + 2/(\beta^2 + \beta)] \approx I_{REF}(1 - 2/\beta^2)$$

- [2] Interpretation.** The copying error collapses from $2/\beta$ to $\approx 2/\beta^2$ — at $\beta = 100$, from 2% to 0.02%, a hundredfold improvement — because the follower's current gain divides the stolen base current by $(\beta+1)$. Also mention the second-order fix left over: Early-effect mismatch (unequal V_{CE}), addressed by cascode/Wilson mirrors ($R_{out} \approx \beta r_o/2$) — one line each.
- ⚠ Mistakes: forgetting BOTH base currents are stolen (factor 2) · connecting Q_3 backwards (its emitter must feed the bases) · claiming compensation fixes the Early effect (it fixes only base-current error).

3.7 The Widlar Current Source ★ REPEATED PYQ TOPIC

PYQs: 2082 Bhadra Q6 [5+2] (operation + expressions + advantage) and 2081 Ashwin Q8 [5] (draw it + derive its output resistance).

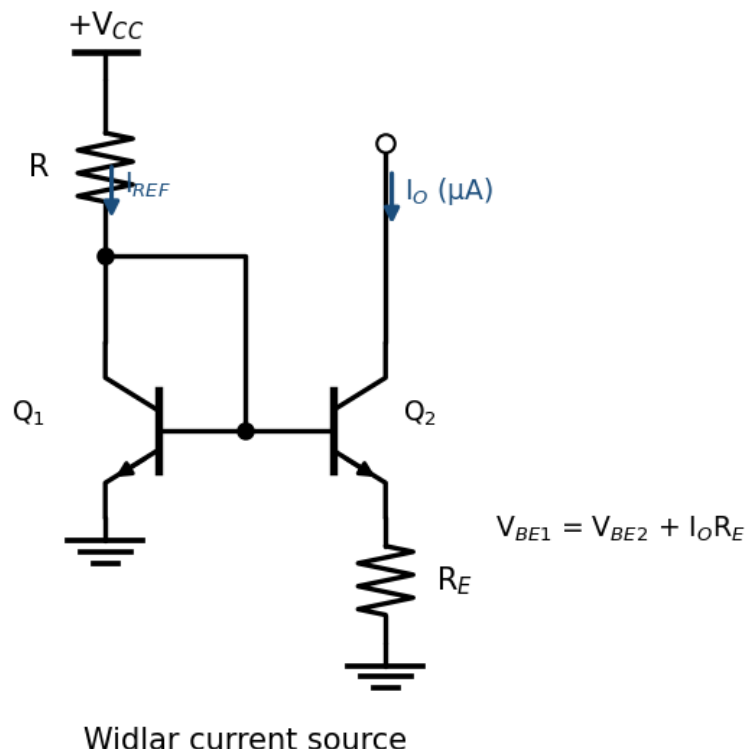


Fig. 11 — Widlar current source: a simple mirror with resistor R_E in the OUTPUT transistor's emitter only. The V_{BE} mismatch it creates is what scales the current down. [rendered locally]

Operation + design-equation derivation [the 5 marks]

- 1. KVL around the two base-emitter loops: $V_{BE1} = V_{BE2} + I_O R_E$ ($I_{E2} \approx I_O$ for large β).
- 2. Both junctions obey $I_C = I_S e^{V_{BE}/V_T}$ with matched I_S , so $V_{BE1} - V_{BE2} = V_T \ln(I_{REF}/I_O)$. Substituting:

$$V_T \ln(I_{REF}/I_O) = I_O R_E \quad \text{(the Widlar design equation — transcendental in } I_O; \text{ solve iteratively, or directly for } R_E)$$

- 3. **Design example (quote it):** $I_{REF} = 1 \text{ mA}$, $I_O = 10 \text{ } \mu\text{A} \Rightarrow R_E = V_T \ln(100)/10 \text{ } \mu\text{A} = (25 \text{ mV} \times 4.605)/10 \text{ } \mu\text{A} \approx \mathbf{11.5 \text{ k}\Omega}$. Making $10 \text{ } \mu\text{A}$ with a plain mirror would need $R \approx (12 - 0.7)/10 \text{ } \mu\text{A} \approx \mathbf{1.13 \text{ M}\Omega}$ — a hundred times the silicon area. That contrast IS the advantage answer.

Advantages over the simple mirror [the 2 marks]

- (1) Generates **microampere currents with kilohm resistors** — IC-area-friendly (the log compresses the ratio).
- (2) R_E is emitter degeneration $\Rightarrow$ far **higher output resistance** (better current source) and reduced sensitivity to V_{BE} mismatch and temperature.

Output-resistance derivation — PYQ 2081 Q8 [5]

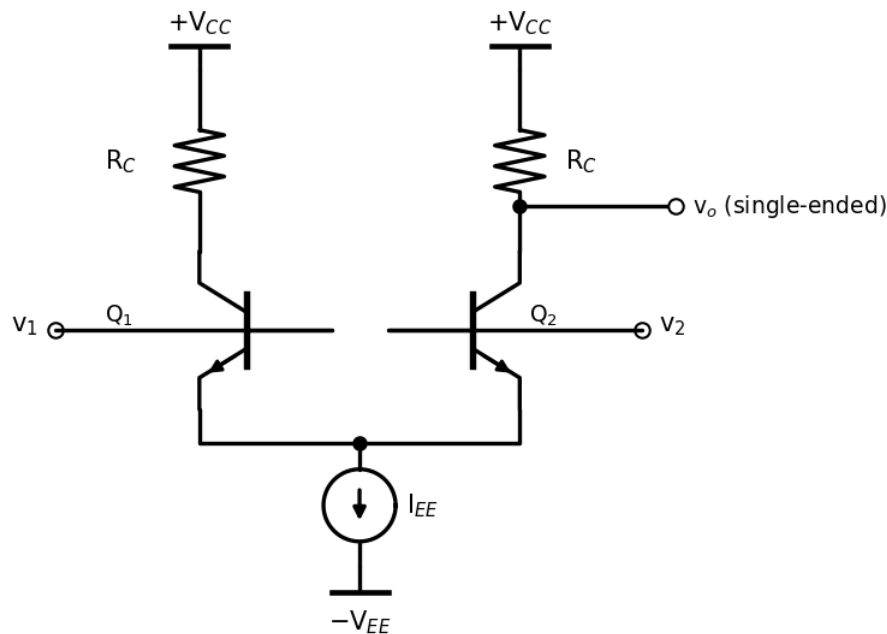
- Setup:** small-signal model of Q_2 ; the base sees the diode-connected Q_1 side — a low impedance $\approx 1/g_{m1}$ — so take the base as AC ground. The emitter then sees $R_E' = R_E \parallel r_{\pi}$ to ground. Apply a test source v_x , i_x at the collector.
- 1. Nearly all of i_x exits through the emitter network (base current is small): $v_e = i_x \cdot R_E'$.
- 2. With the base grounded, $v_{\pi} = -v_e$, so the controlled source $g_m v_{\pi} = -g_m v_e$ opposes i_x , forcing the difference through r_o .

- **3.** KCL at the collector: $i_x = g_m v_{\pi} + (v_x - v_e)/r_o \Rightarrow v_x = r_o(i_x + g_m v_e) + v_e = r_o i_x(1 + g_m R_E') + i_x R_E'$.

$$R_{out} = v_x/i_x = r_o(1 + g_m R_E') + R_E' \approx r_o[1 + g_m(R_E \parallel r_{\pi})] \gg r_o$$

- $\triangle$ Mistakes: putting R_E under the *reference* transistor · dropping the I_n (writing a linear ratio) · claiming $R_{out} = r_o + R_E$ (misses the $g_m R_E'$ multiplication that does all the work).

3.8 The Differential Amplifier



signal current $\pm g_m v_d/2$ per side; output uses ONE side only

Fig. 12 — Emitter-coupled pair over a tail current source, resistive (passive) loads. A differential input v_d splits I_{EE} : signal current $\pm g_m v_d/2$ in each collector. [rendered locally]

- **Operation:** the tail source fixes $I_{E1} + I_{E2} = I_{EE}$; a differential input $v_d = v_1 - v_2$ tilts the split. For small v_d each device carries $I_{EE}/2 \pm g_m v_d/2$ with $g_m = (I_{EE}/2)/V_T$.
- **Gains:** differential (double-ended) $A_d = g_m R_C$; **single-ended $A_d = g_m R_C/2$** — half the signal current is thrown away in the unused collector (remember this for §3.9). Common-mode: $A_{cm} \approx -R_C/(2R_{EE}) \rightarrow \mathbf{CMRR} = \mathbf{A_d/A_{cm}}$, huge when the tail is a good current source (Widlar!).

3.9 Active Loads — PYQ 2082 Baishakh Q8 [5]: show the gain with an active load is TWICE that with a passive load

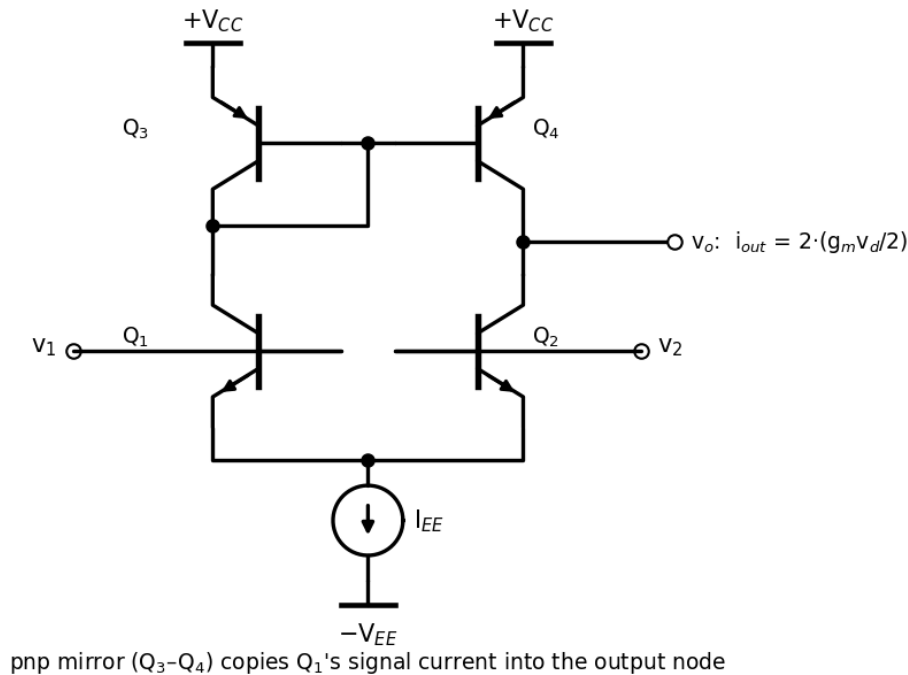


Fig. 13 — Differential pair with a pnp current-mirror ACTIVE load: Q₃ (diode-connected) senses Q₁'s signal current and Q₄ re-injects a copy into the single-ended output node. [rendered locally]

The proof (four numbered steps — this exact chain is the 5 marks)

- **1.** A differential input creates equal and opposite signal currents: $i_1 = +g_m v_d / 2$ in Q₁, $i_2 = -g_m v_d / 2$ in Q₂ (i.e. $g_m v_d / 2$ flowing OUT of the output node).
- **2. Passive load, single-ended output:** only Q₂'s current reaches the output resistor: $|v_o| = (g_m v_d / 2) \cdot R_L \Rightarrow |A_{v, \text{passive}}| = g_m R_L / 2$. Q₁'s half is wasted.
- **3. Active (mirror) load:** Q₁'s current i_1 flows through diode-connected Q₃; Q₄ mirrors it and **sources the same $g_m v_d / 2$ INTO the output node**, where Q₂ is simultaneously sinking $g_m v_d / 2$. KCL: $i_{\text{out}} = g_m v_d / 2 + g_m v_d / 2 = g_m v_d$.
- **4.** Same load resistance, twice the output current $\Rightarrow |A_{v, \text{active}}| = g_m R_L = 2 \times |A_{v, \text{passive}}|$ ■ — the mirror recovers the wasted half. Bonus line: the actual node resistance is $r_{o2} \parallel r_{o4}$, so practically $A_v = g_m (r_{o2} \parallel r_{o4})$ — gains in the thousands with almost no DC voltage drop, which is WHY ICs use current sources instead of resistors as loads.
- ⚠ Mistake: arguing 'gain doubles because r_o is large' — the doubling is the recombined signal current; the high node resistance is a separate (additional) benefit.

3.10 Output Stages (bridge to Chapter 4)

● **LOW** An op-amp's final stage must drive low-impedance loads with watts, low distortion and decent efficiency: source/emitter followers and complementary Class-AB pairs. The full treatment — classes A/B/AB, efficiencies 25/50/78.5%, crossover distortion — is Chapter 4 §4.1–4.5; from this chapter just carry the *why*: low Z_{out} and current capability.

Chapter 3 Final Revision

Conceptual summary (read this the night before)

Positive feedback with unity complex loop gain ($|A\beta| = 1$, $\angle A\beta = 0^\circ$ — Barkhausen) turns an amplifier into a sinusoidal oscillator; the frequency-selective β network picks f_0 and charges an entry fee the amplifier must repay: three-section RC ladder ($f_0 = 1/2\pi RC\sqrt{6}$, fee 29), Wien bridge ($f_0 = 1/2\pi RC$, fee 3, $R_3 = 2R_4$), Colpitts (capacitive divider, $C_{eq} = C_1C_2/(C_1+C_2)$), Hartley (tapped coil, $L_{eq} = L_1+L_2+2M$), crystal for ppm stability. The 555 relaxes a capacitor between $1/3$ and $2/3$ of V_{CC} : $t_H = 0.693(R_A+R_B)C$, $t_L = 0.693R_BC$, $f = 1.44/(R_A+2R_B)C$, supply-independent. IC biasing copies one reference current: simple mirror errs by $2/\beta$, a compensating follower shrinks it to $2/\beta^2$, and Widlar's emitter resistor buys microamp outputs ($V_T \ln(I_{REF}/I_O) = I_O R_E$) plus $R_{out} \approx r_o(1+g_m R_E)$. The differential pair splits $\pm g_m v_d/2$ per side; a mirror load recombines both halves — doubling single-ended gain to $g_m(r_{o2} \parallel r_{o4})$.

Important diagrams checklist

Figure	Priority	Marked in PYQ as
Fig. 4 — Wien bridge circuit	● HIGH	2083 Q5 ★
Fig. 2 — RC phase-shift circuit	● HIGH	2082 Bhadra Q5 ★
Fig. 6 — Colpitts circuit	● HIGH	2081 Q6 ★
Fig. 7 — Hartley circuit	● HIGH	2082 Baishakh Q6 ★
Fig. 8 + 9 — 555 astable + waveforms	● HIGH	2081 Q7, 2082 Ba Q11 ★
Fig. 10 — simple & compensated mirrors	● HIGH	2083 Q6
Fig. 11 — Widlar source	● HIGH	2082 Bh Q6, 2081 Q8 ★
Fig. 12 + 13 — diff pair, passive & active load	● MEDIUM	2082 Ba Q8
Fig. 1 — feedback loop block diagram	● MEDIUM	Barkhausen lead-ins
Crystal equivalent circuit	● MEDIUM	fresh-question candidate

Formula sheet

Barkhausen: $|A\beta| = 1$, $\angle A\beta = 0^\circ$ · **RC-PS:** $f_0 = 1/(2\pi RC\sqrt{6})$, $|A| \geq 29$ · **Wien:** $f_0 = 1/(2\pi RC)$, $A = 3$ ($R_3 = 2R_4$), $\beta_0 = 1/3$

Colpitts: $f_0 = 1/(2\pi\sqrt{LC_{eq}})$, $C_{eq} = C_1C_2/(C_1+C_2)$, $|A| \geq C_2/C_1$ · **Hartley:** $f_0 = 1/(2\pi\sqrt{L_{eq}C})$, $L_{eq} = L_1+L_2+2M$, $|A| \geq L_1/L_2$

555: $t_H = 0.693(R_A+R_B)C$ · $t_L = 0.693R_BC$ · $T = 0.693(R_A+2R_B)C$ · $f = 1.44/[(R_A+2R_B)C]$ · $D = (R_A+R_B)/(R_A+2R_B)$

Mirrors: $I_O = I_{REF}/(1+2/\beta) \rightarrow I_{REF}/(1+2/\beta^2)$ · **Widlar:** $V_T \ln(I_{REF}/I_O) = I_O R_E$, $R_{out} \approx r_o(1+g_m(R_E \parallel r_{\pi}))$

Diff pair: $A_d(\text{se, passive}) = g_m R_C/2$ · $A_d(\text{active}) = g_m(r_{o2} \parallel r_{o4})$ · **CMRR** = A_d/A_{cm} · $g_m = (I_{EE}/2)/V_T$

Priorities / mistakes / derivation list

- **HIGH** All four oscillators with full frequency derivations (★4/4) · mirror error $2/\beta \rightarrow 2/\beta^2$ · Widlar equation + R_{out} (★3/4) · 555 derivation (2/4).
- **MEDIUM** Active-load $\times 2$ proof · crystal oscillator · CMRR discussion · op-amp square/triangle generator.
- **LOW** Precision rectifiers · cascode/Wilson mirrors · output-stage bridge.
- ⚠ Top mistakes: $|A\beta| > 1$ quoted as steady state · C_{eq}/L_{eq} series-parallel swaps · discharge path includes R_A (it doesn't) · Widlar R_E on the wrong transistor · 'active load doubles gain because r_o is big' · forgetting the amplifier's own 180° in the phase-shift count.

- Derivations to rehearse on paper: Wien β algebra $\rightarrow f_0$, $A = 3$ · ladder phase condition $\rightarrow \sqrt{6}$ and 29 · both LC resonance sums · 555 t_H/t_L · mirror KCL chains · Widlar KVL + R_{out} test-source · active-load KCL.

Complete Chapter 3 PYQ Bank (wording preserved)

Year / Exam	Q	Question (condensed, wording preserved)	Mark s	Type
2083 Baishakh (Back)	5	What is the main difference between RC and LC oscillator circuits in terms of application? Explain the operation of Wien Bridge Oscillator with circuit diagram and necessary mathematical expressions.	1+5	Theory + Derivation
2083 Baishakh (Back)	6	Explain how a base current compensation circuit reduces the difference between I_O and I_{REF} in a simple current mirror circuit.	5+2	Derivation
2082 Bhadra (Regular)	5	State Barkhausen criteria for sinusoidal oscillation. Explain the working principle of RC phase shift oscillator with necessary expression and circuit diagram.	6	Theory + Derivation
2082 Bhadra (Regular)	6	With a neat circuit diagram and necessary mathematical expressions, explain the operation of Widlar current source. What is the advantage of Widlar current source over simple current source?	5+2	Theory + Derivation
2082 Baishakh (Back)	6	Draw the circuit diagram of LC Hartley oscillator. Derive its frequency of oscillation.	4	Derivation
2082 Baishakh (Back)	8	Show that the voltage gain of the differential amplifier with active load is twice that with passive load.	5	Proof
2082 Baishakh (Back)	11	Derive the frequency of oscillation of 555 timer Astable Multivibrator.	4	Derivation
2081 Ashwin (Regular)	6	State Barkhausen criterion for sinusoidal oscillation. Draw the circuit diagram of Colpitts oscillator and write its frequency of oscillation.	4	Theory
2081 Ashwin (Regular)	7	Derive the frequency of oscillation of 555 timer Astable Multivibrator.	4	Derivation
2081 Ashwin (Regular)	8	Draw a Widlar Current Source. Derive an expression for the output resistance of Widlar Current Source.	5	Derivation

PYQ → What to Study Mapping

PYQ	Topic	Notes section(s)	Required skill
2083 Q5 [1+5]	RC-vs-LC + Wien	§3.2-B (Figs. 4-5) + §3.1	application contrast; β algebra $\rightarrow f_0 = 1/2\pi RC$, $A = 3$
2083 Q6 [5+2]	Compensated mirror	§3.6 (Fig. 10)	KCL error chains $2/\beta \rightarrow 2/\beta^2$ + interpretation
2082 Bh Q5 [6]	Barkhausen + RC-PS	§3.1 + §3.2-A (Figs. 2-3)	criterion + $60^\circ \times 3$ story + f_0 and $ A \geq 29$
2082 Bh Q6 [5+2]	Widlar	§3.7 (Fig. 11)	KVL $\rightarrow$ design equation + design example + 2 advantages
2082 Ba Q6 [4]	Hartley	§3.3-B (Fig. 7)	circuit + $L_{eq} = L_1 + L_2 + 2M \rightarrow f_0$
2082 Ba Q8 [5]	Active-load $\times 2$	§3.8 + §3.9 (Figs. 12-13)	four-step signal-current recombination proof
2082 Ba Q11 [4]	555	§3.4 (Figs. 8-9)	exponential-law derivation of t_H , t_L , f
2081 Q6 [4]	Barkhausen + Colpitts	§3.1 + §3.3-A (Fig. 6)	criterion + circuit + f_0 with C_{eq}
2081 Q7 [4]	555	§3.4	same derivation, word-for-word repeat
2081 Q8 [5]	Widlar R_{out}	§3.7 (R_{out} part)	test-source derivation $R_{out} \approx r_o(1 + g_m R_E)$

Final Coverage Audit

- ✓ Syllabus 3.1–3.10 all present, in order; never-asked items (square/triangle generators, crystal, precision rectifiers, output-stage bridge) included at readiness depth with priority labels.

- ✓ References honoured: Sedra–Smith (loop-gain method, mirror error analysis, Widlar R_{out} , active-load proof), Boylestad/Floyd (oscillator circuit forms, 555 internals), Millman (feedback framing). Notation unified: $V_T \approx 25 \text{ mV}$, $\beta \equiv h_{FE}$, $\alpha = 1/\omega RC$ flagged where used.
- ✓ All 10 PYQs embedded in their sections AND repeated in the bank + mapping; every demanded derivation written out in numbered steps (Wien algebra, ladder conditions, both LC resonances, 555 exponentials, both mirror error chains, Widlar design equation and R_{out} , active-load $\times 2$).
- ✓ ★ Repeated topics flagged (oscillator 4/4, current source 3/4, 555 2/4); ● priorities from PYQ evidence; 13 figures rendered locally — the response curves (Figs. 3, 5, 9) computed directly from the network equations, per the fetch report (0/13 from web).

End of Chapter 3 notes. Companions: Chapter 1 (BJT, markdown), Chapter 2 (FET, PDF — same series), Chapters 4–5 compact notes pending the same full-depth upgrade.